Pain Detection via Knowledge Distillation

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1 Introduction

Automated pain detection is critical for clinical settings where patients cannot verbally communicate, such as infants, individuals with cognitive impairments, or unconscious patients. El Morabit & Rivenq (2022) proposed a DeiT-based solution with ResNet-50 as teacher network, achieving 84.15% accuracy on UNBC-McMaster dataset but with limited exploration of alternative teacher architectures. This study investigates whether different teacher network architectures—specifically EfficientNet-B3 and Swin Transformer—can provide similar or superior accuracy with better knowledge transfer efficiency. Using the SynPAIN dataset, we aim to evaluate whether architectural alignment between teacher and student networks improves distillation performance for pain detection tasks.

2 Research Question

How do different teacher network architectures (ResNet-50, EfficientNet-B3, Swin Transformer) compare in terms of accuracy and knowledge transfer efficiency when used for distilling knowledge to DeiT for automated pain detection from facial expressions?

3 Methodology

Model Selection: Fine-tune Data-efficient Image Transformer (DeiT) with three different teacher architectures: ResNet-50 (baseline CNN), EfficientNet-B3 (efficient CNN with compound scaling), and Swin Transformer (hierarchical attention-based architecture)

Dataset: SynPAIN synthetic facial expression dataset containing balanced pain/no-pain classes with diverse pain intensity expressions for robust binary classification evaluation

Preprocessing Pipeline: Face detection using MTCNN, alignment and cropping to 256×256 pixels, data augmentation during training phase

Training Configuration: Train models using patch size 32×32, learning rate 1e-5, 30 epochs, batch size 64, Adam optimizer with dual loss function (Binary Cross-Entropy + Hard Distillation Loss)

Performance Metrics: Evaluate accuracy, precision, recall, F1-score, model parameter count, model size (MB), and training convergence rate to assess real-time feasibility for clinical pain monitoring applications

Statistical Validation: 5-fold cross-validation with paired t-tests to ensure robust performance comparison across teacher architectures

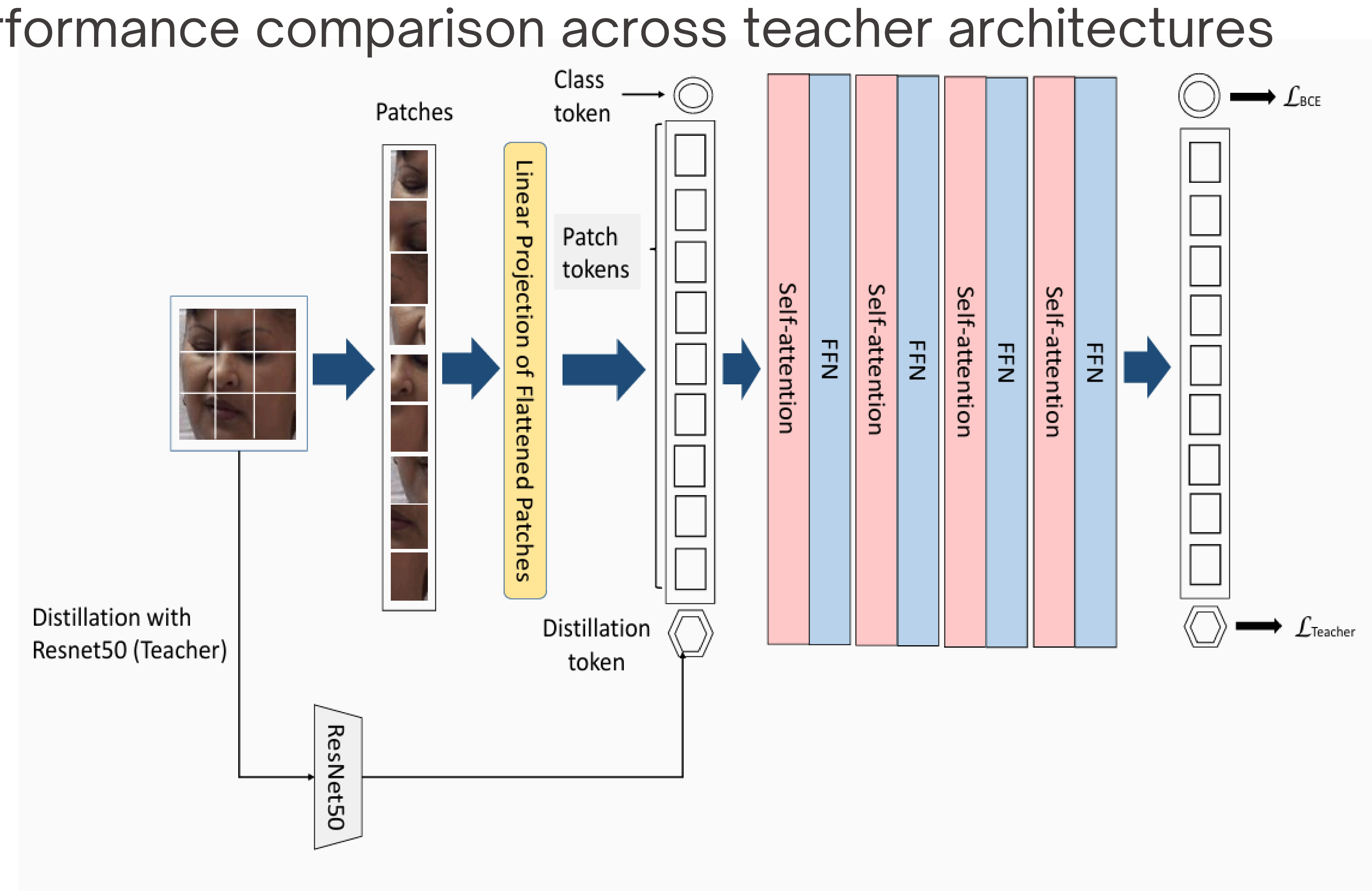

Fig. 2. An overview of the proposed pipeline for Pain Recognition using Data-efficient image transformer (Deit) [6]. (FFN stands for Feed Forward Network. BCE is the Binary Cross Entropy.)

References

S. El Morabit and A. Rivenq, "Pain Detection From Facial Expressions Based on Transformers and Distillation," in 2022 11th International Symposium on Signal, Image, Video and Communications (ISIVC), Valenciennes, France, 2022, pp. 1-5, doi: 10.1109/ISIVC54825.2022.9800746.

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